Culture: For inoculation, it is best to use mucus flakes (if present in the specimen) on MacConkey agar and SS agar. After overnight incubation at 37°C, the plates are observed for characteristic colonies, which is confirmed by Grams staining and biochemical reactions.

7.8.7 Treatment and Prevention

- Uncomplicated shigellosis is a self – limiting condition that usually recovers spontaneously.
- In acute cases, oral rehydration therapy (ORT) is done.
- In all severe cases, the choice of antibiotic should be based on the sensitivity of prevailing strain.
- Many strains are sensitive to Nalidixic acid and Norfloxacin.
- Improving personal and environmental sanitation.
- The detection and treatment of patients and carriers.

7.9 *Salmonella Typhi* (Eberthella Typhi)

The genus *Salmonella* consists of bacilli that parasites the intestines of vertebrates and human beings. It causes **Enteric fever**, which includes **Typhoid** and **Paratyphoid fever**. The most important species of the genus is *Salmonella typhi* which causes typhoid fever.

7.9.1 Morphology

Salmonellae are Gram – negative rods (1–3µm× 0.5 µm in size). They are motile with peritrichous flagella, non – capsulated and non – sporulated (Figure 7.14).

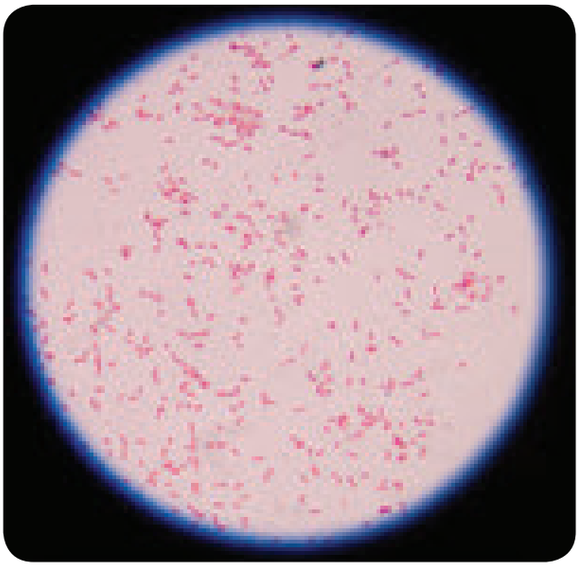


Figure 7.14: Gram staining of *Salmonella typhi*

7.9.2 Cultural Characteristics

They are aerobic and facultative anaerobe, optimum temperature - 37°C and pH is 7–7.5.

They grow on the following media and show the following characteristic colony morphology (Table 7.12).

Table 7.12: Colony morphology of *Salmonella typhi*

Media	Colony Morphology
Nutrient Agar	Colonies are large, circular, smooth, translucent
MacConkey Agar	Colourless colonies (non – lactose fermenters)
SS – Agar	Colourless colonies with black centered.

7.9.3 Pathogenicity

Salmonella typhi causes typhoid fever and its pathogenesis is discussed in flowchart 7.6.

Source of infection – food, feces, fingers, flies

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